



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 07



FOTON FOUR-WHEEL TRACTOR
Model : TD904

FOTON TD904 FOUR-WHEEL TRACTOR

Test Report No	:	2013 - 07
Date of Test	:	January 3, 25, & February 4, 2013
Test Valid Until	:	February 4, 2018
Brand	:	FOTON
Model	:	TD904
Serial No.	:	FTTD54AOLCN001032
Type	:	Four-Wheel Drive Tractor
Manufacturer	:	FOTON LOVOL INTERNATIONAL HEAVY INDUSTRIES CO., LTD. No. 192, S. Beihai road, Weifang, Shandong, P.R.O. China
Test Requested by	:	GREEN VALLEY MACHINERIES Gensan Drive, Koronadal City, South Cotabato
Machine Classification	:	Commercial

SUMMARY

The FOTON TD904 Four-Wheel tractor was tested to establish its performance. It was tested at the original settings set by the manufacturer.

The Maximum Power developed at the Power-Take-Off (PTO) shaft of the tractor was 60.7 kW at 561 rpm PTO Shaft Speed. The Engine Crankshaft Speed at this point was 2043 rpm with Fuel Consumption Rate of 20.1 L/h. The PTO Shaft Torque at this point was 105.4 kg-m. The Maximum PTO Shaft Torque was 126.0 kg-m which occurred at 432 rpm PTO Shaft Speed.

The Radii of Turning Circle of the tractor when making a right turn were 5275 mm without brake and 4390 mm when brake was applied on the right rear wheel. On the other hand, the Radii of Turning Circle of the tractor when making a left turn were 5188 mm without brake and 4438 mm when brake was applied on the left rear wheel.

The Radii of Turning Area of the tractor when making a right turn were 5315 mm without brake and 4430 mm when brake was applied on the right rear wheel. On the other hand, the Radii of Turning Area of the tractor when making a left turn were 5228 mm without brake and 4478 mm when brake was applied on the left rear wheel.

The Travelling Speeds of the tractor at the engine full throttle speed setting (Pre-set) recommended by the manufacturer/Applicant of test at different gear settings were the following: At “LOW” speed range the travelling Speeds were 2.35 kph, 3.67 kph, 5.89 kph and 8.08 kph at 1st, 2nd, 3rd, and 4th gear settings, respectively, while at “HIGH” speed range the travelling Speeds were 9.96 kph, 15.6 kph, 24.8 kph and 34.4 kph at 1st, 2nd, 3rd, and 4th gear settings, respectively. At “REVERSE” speed range the travelling speeds were 3.21 kph, 4.87 kph, 7.93 kph and 10.8 kph at 1st, 2nd, 3rd and 4th gear settings, respectively.

SCOPE OF TEST

The tractor was tested to establish its performance characteristics. The test was conducted on January 3, 25, and February 4, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna. It was evaluated on the following items:

1. Suitability of operator's manual;
2. Verification of Specifications
3. Laboratory tests
 - a. PTO load performance test;
 - b. Determination of Turning Circle
 - c. Determination of Turning Area
 - d. Determination of Travelling Speeds

METHOD OF TEST

The tractor was tested based on Philippine Agricultural Engineering Standards, PAES 119:2001 Agricultural Machinery – Four-Wheel Tractor – Methods of Test.

DESCRIPTION OF THE TRACTOR

The FOTON TD904 Four-Wheel Tractor (**Figure 1**) has a PTO rated power of 59.4 kW (79.6 Hp) four-wheel drive tractor as stated in the operator's manual.

The tractor used a four-stroke, water-cooled, turbocharged, four cylinder diesel engine. Fuel feed system is a force-feed type using a circulation pump. There were two fuel tanks (**Figure 2**) located at the bottom of the tractor's body just beneath tractor's steps. It utilized an Oil-bath type air cleaner with pre-cleaner on top of it(**Figure 3**). Lubrication system was a force-feed lubricating type. Cooling was done through circulation of water from cylinder jacket to the radiator using a circulation pump.

It was started by an electric starting system. A vertical muffler was used in the exhaust system.

The tractor's transmission system had eight forward speeds and four reverse speeds, which were grouped into three ranges. The speed ranges were "LOW", "HIGH", and "REVERSE" while the gear settings were 1st, 2nd, 3rd, and 4th. The main gear was fully synchronized while the range gear could only be shifted when the tractor is completely stopped. The range and main gear settings could be set through levers at the right side of operator's seat. A parking brake was also provided and can be actuated through a lever at the left side of the operator's seat.

The tractor's direction was being controlled through a hydrostatic power steering system with the steering wheel (**Figure 4**) located in front of the driver's seat. The brakes of the rear wheels were independent of each wheel, however, they could be locked-on through the pedals. A differential lock could be set by foot pedals located on the tractor's flooring just underneath the operator's seat. The size of the rear wheel tires were 43.0 cm x 86.4 cm (16.9 in x 34.0 in) and the front wheel tires were 34.5 cm x 61.0 cm (13.6 in x 24.0 in). Both have spikes.

The tractor had a three-point hitch linkage (**Figure 5**) for mounted implements. It had a fixed (**Figure 6a**) and swinging drawbars (**Figure 6b**) for pulling a trailer or any trailing implement.

The Power-Take-Off shaft (**Figure 6c**) of the tractor was located at the rear of the tractor and was set at 540 rpm. The PTO shaft could be engaged/disengaged through a lever located at the left side of the operator seat.

The tractor had two headlights, two brake lights, four turning lights and two working lights.

The operator's seat (**Figure 7**) was adjustable for operator's comfort.

The tractor controls (**Figure 8**) were located in front of and beside the operator's seat and included the following: steering wheel, clutch pedal, hand throttle lever, ignition key, main gear shift lever, range gear shift lever, differential lock pedals, brake pedals, foot throttle pedal, PTO controls, hydraulic lift position control lever and auxiliary hydraulic control lever. The panel board (**Figure 8**) which was also located in front of the operator's seat contained the following: tachometer, fuel gage, hour meter and coolant temperature gage. **Table 1** shows the detailed specifications of the tractor.



Figure 1. FOTON TD904 Four-Wheel Tractor

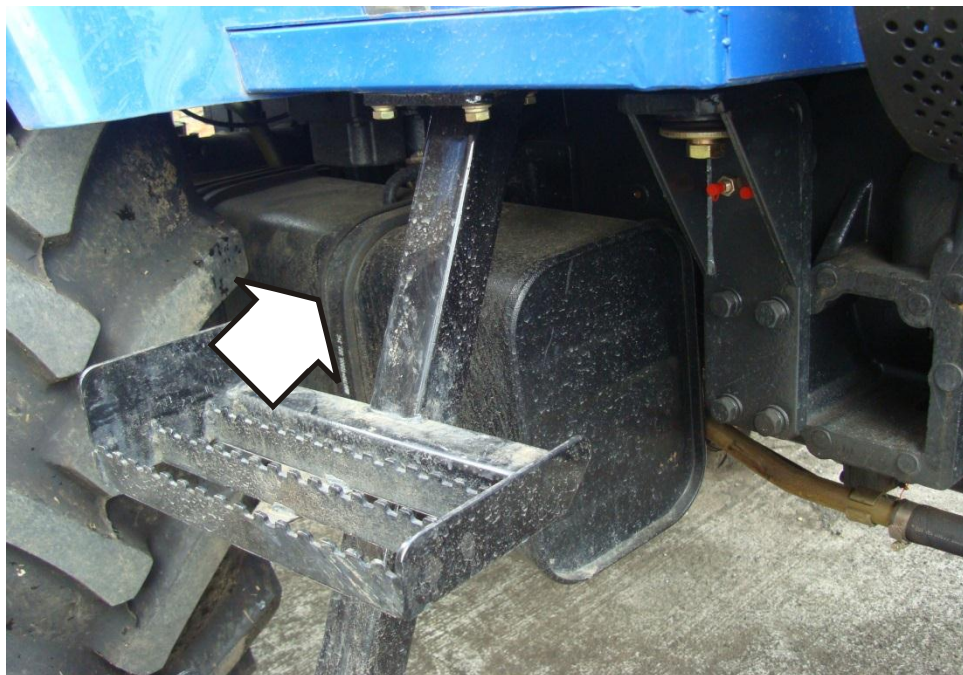


Figure 2. Fuel tanks



Figure 3. Air Cleaner



Figure 4. Steering wheel

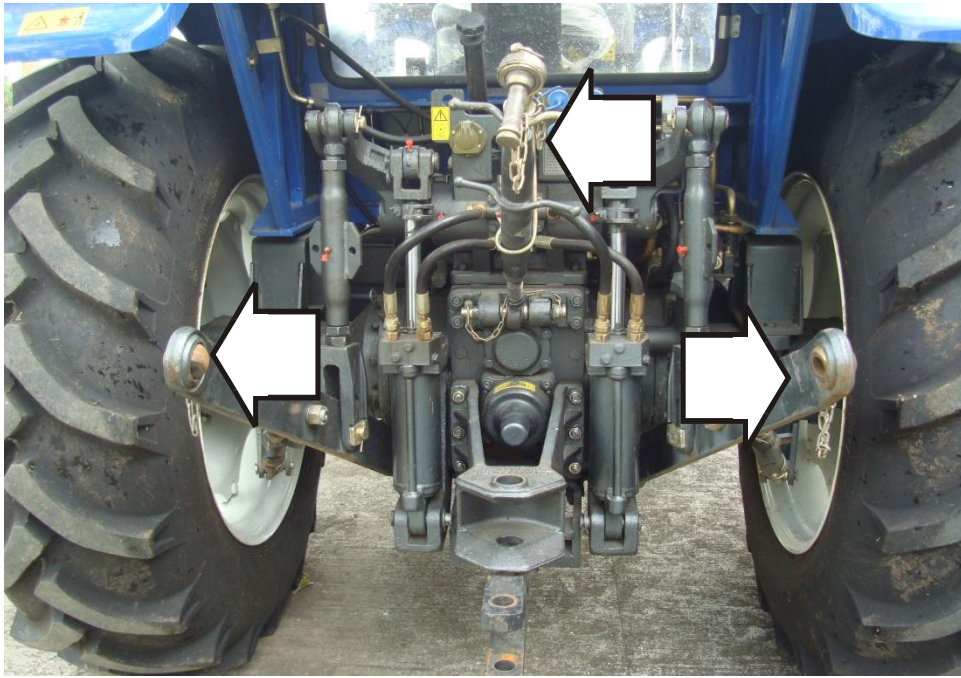


Figure 5. Three-Point Hitch

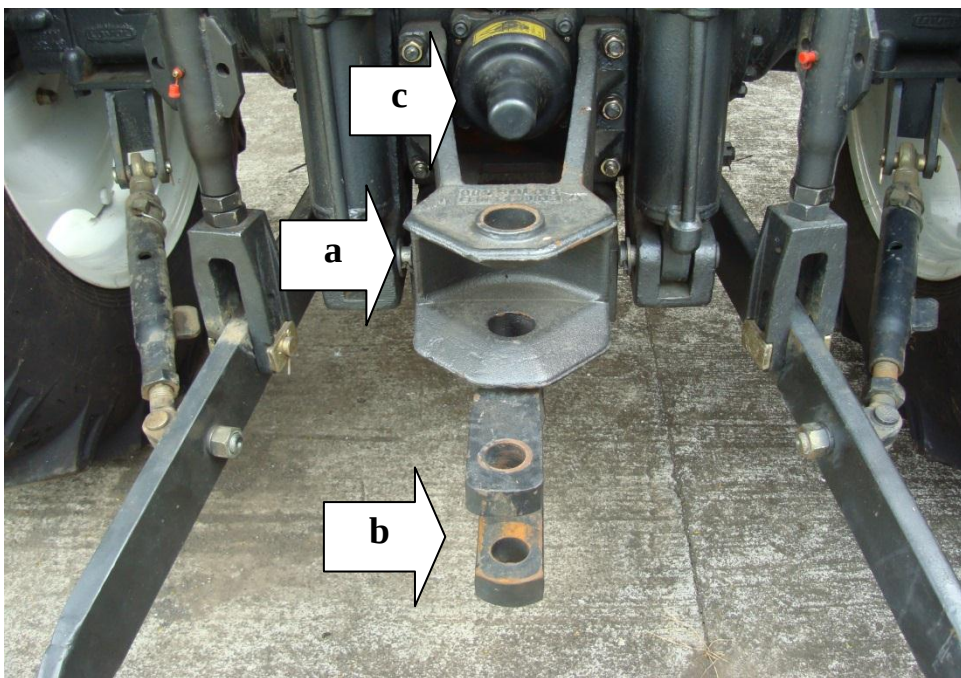


Figure 6. (a) Fixed drawbar
(b) Swinging drawbar
(c) PTO output shaft



Figure 7. Operator's seat

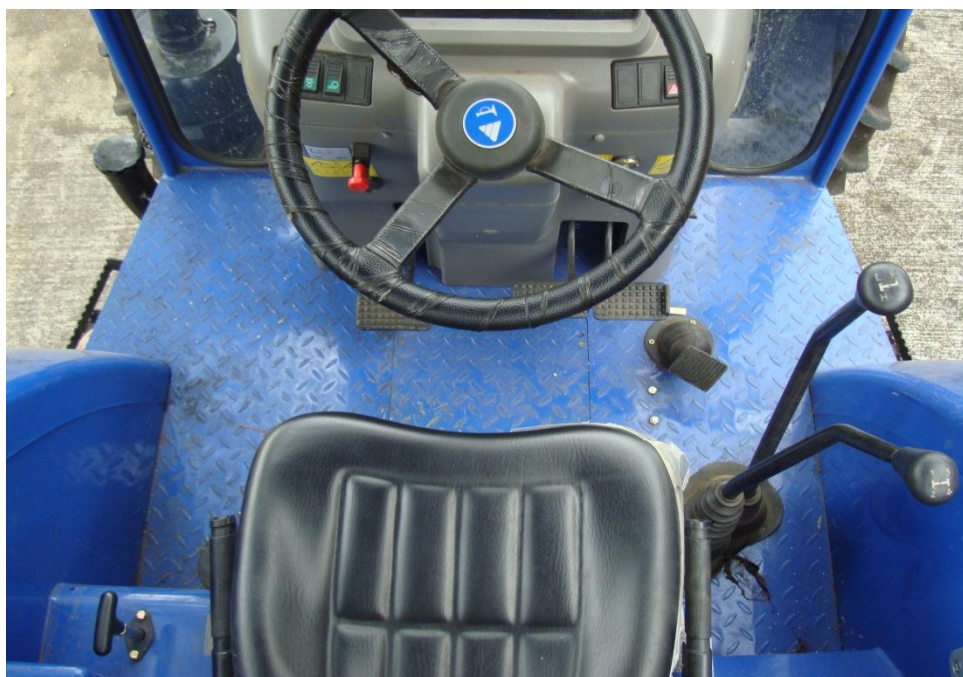


Figure 8. Tractor's control and panel board

SPECIFICATIONS

Table 1. Detailed specifications of the FOTON TD904 Four-Wheel Tractor

1.	Tractor specifications (Based on actual ocular inspection of the tractor and operator's manual submitted)		
2.1	Brand	:	FOTON
2.2	Model	:	TD90D
2.3	Serial No.	:	FTTD54AOLCN001032
2.4	Make	:	FOTON LOVOL INTERNATIONAL HEAVY INDUSTRY CO., LTD., China
2.5	Type	:	Four-Wheel Drive
2.6	Rated Maximum PTO Power (kW)	:	59.4 (Based on Operator's manual)
2.	Dimensions and Weight		
1.1	Overall length (mm) With front ballast	:	4451
1.2	Overall width (mm)	:	2072
1.3	Overall height (mm)	:	2900 (Canopy)
3.	Tractor's engine		
3.1	Brand	:	FOTON
3.2	Type	:	Four-Stroke Cycle, Water-cooled, Turbocharged, Four Cylinder Diesel Engine
3.3	Model	:	1004C-P4TRT90
3.4	Make	:	Tianjin Lovol Engines Co., LTD., China
3.5	Serial Number	:	HC516889XN01
3.6	Rated Output Power (kW)	:	66.2
3.7	Number of Cylinder	:	4

Table 1. Con't

3.8	Bore x stroke (mm)	:	100 x 127
3.9	Total displacement volume (cc)	:	3989.8
3.10	Fuel system		
3.10.1	Type of fuel feed	:	Circulation pump
3.10.2	Fuel used	:	Diesel
3.11	Cooling system	:	Water-cooled
3.12	Lubricating system	:	Force-feed type
3.13	Starting system	:	Electric type using battery
3.14	Air cleaner	:	Oil-bath type (with pre-cleaner)
3.15	Exhaust system	:	Vertical muffler
4.	Power-Take-off		
4.1	Location	:	At the rear of the tractor
4.2	Diameter of PTO shaft end (mm)	:	35.0
4.3	Number of splines	:	6
4.4	Height above ground (mm)	:	740
4.5	Rated Shaft Speed(rpm)	:	540
4.6	Direction of rotation (viewed from the rear of the tractor)	:	Clockwise
4.7	Mode of Operation	:	Lever
5.	Drawbar		
5.1	Type	:	Swinging and fixed
5.2	Diameter of hitchpoint (mm)		
a.	Swinging type drawbar	:	33.0
b.	Fixed	:	43.0
6.	Tire		
6.1	Front		
6.1.1	Number	:	2
6.1.2	Size (cm)	:	34.5 x 61.0
6.1.3	Track width (mm)	:	1625

Table 1. Con't

6.2	Rear		
6.2.1	Number	:	2
6.2.2	Size	:	43.0 x 86.4
6.2.3	Track width (mm)	:	1626
7.	Wheel base (mm)	:	2400
8.	Ground Clearance (mm)	:	402 (Drawbar bracket)

Table 2. Dimension of linkage geometry of FOTON TD904 tractor
(see also **Figure 9**)

Particulars		Data (mm)
1. Length of Lift Arms	A	230
2. Length of Lower Links	B	902
3. Horizontal distance between two Lower Link Points	u	905
4. Horizontal distance between the two Lift Arm End Points	v	606
5. Distance of Lower Link Pivot Points to Lift Rod Pivot Points on Lower Links	D	370
6. Length of Lift Rods	L	601 - 804
7. Height of Lower Hitch Points relative to the rear wheel axis		
a. Low Position	h	445
b. High Position	H	250
8. Height above ground of Lower hitch points when locked in transport position		997
9. Three-point-hitch		Category II

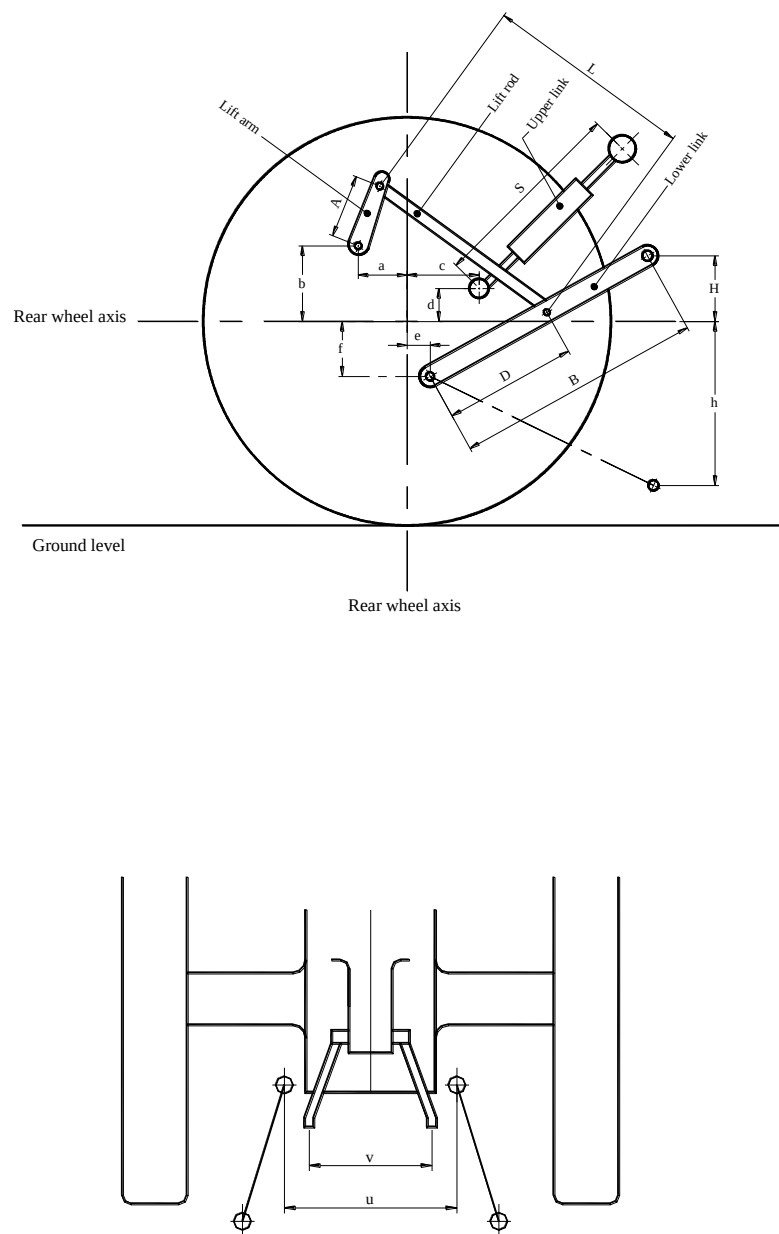


Figure 9. Linkage Geometry

OPERATOR'S MANUAL

The operator's manual submitted with the tractor was presented in English and contained the following: Safety precautions; Operator instruction (Detailed pre-operation and operation procedures); Accessories, spare parts, and consumables; Instruction on maintenance; Storage, Delivery, acceptance and transportation; Technical specifications; Disassembly and disposal; Warranty term; and Appendix.

RESULTS AND DISCUSSION

The FOTON TD904 Four-Wheel tractor was tested to establish its performance. It was tested at the original settings set by the manufacturer.

A. DETERMINATION OF TURNING CIRCLE AND TURNING AREA

The tractor was tested in a suitable concrete area. It was tested by operating the tractor at the slowest speed possible. The steering wheel was rotated to the maximum that it can rotate during turning to the left and right directions. The Radii of Turning Circle and Turning Area with and without brakes were measured. The Radius of Turning Circle is the Radius of the smallest circle tangentially described by the median plane of the outermost wheel, while the Radius of Turning Area is the radius of the smallest circle described by the outermost point of the tractor. Results of the test are shown in **Tables 3** and **4**.

TURNING CIRCLE AND TURNING AREA

Tractor on Test : FOTON
 Model : TD904
 Date of Test : January 3, 2013

Table 3. The Radius of Turning Circle of the tractor

	RIGHT TURN	LEFT TURN
WITHOUT BRAKES	5275	5188
WITH BRAKES	4390	4438

Table 4. The Radius of Turning Area of the tractor

	RIGHT TURN	LEFT TURN
WITHOUT BRAKES	5315	5228
WITH BRAKES	4430	4478

B. DETERMINATION OF TRAVELLING SPEEDS

The Travelling Speeds of the tractor were measured in actual condition. These were measured on a 50-meter concrete road. The tractor was operated at full throttle speed setting and made to run at different range and speed settings. The time elapsed for the tractor to cross the 50-meter distance was obtained. The fourth gear setting at range “HIGH”, yielded the fastest Forward Travelling speed of 34.4 kph, while the first gear setting at range “LOW” was slowest at 2.35 kph. For the Reverse Travelling Speed, the speed at first gear setting was 3.21 kph, while at fourth gear setting the speed was 10.8 kph. In general, the tractor speeds increased with increased range and gear settings. **Table 5** shows the results of the test.

TRAVELLING SPEEDS

Tractor on Test : FOTON
 Model : TD904
 Date of Test : January 3, 2013

Table 5. The Travelling Speeds of the Tractor in kph at different settings

RANGE GEAR SETTINGS	MAIN GEAR SETTINGS (kph)			
	FIRST	SECOND	THIRD	FOURTH
LOW	2.35	3.67	5.89	8.08
HIGH	9.96	15.6	24.8	34.4
REVERSE	3.21	4.87	7.93	10.8

C. PTO PERFORMANCE LOAD TEST

The power at the Power-Take-Off shaft of the tractor was obtained using a Portable Waterbrake Dynamometer. It was tested at full throttle speed setting of the tractor. Data such as dynamometer load, PTO rpm, Engine rpm, fuel consumption were obtained at 20 & 40 rpm speed increments. The Maximum No-Load speed of the engine was 2305 rpm. The Maximum PTO Output Power obtained was 60.7 kW at 561 rpm PTO shaft speed. At this point, the PTO torque was 105.4 kg-m with tractor's Fuel Consumption Rate of 20.1 L/h. No breakdown and other malfunctioning occurred during the test. The test set-up is shown in **Figure 10** and the results of test are shown in **Table 6** and plotted in **Figure 11**.



Figure 10. The Tractor on the PTO test set-up

PTO PERFORMANCE TEST

Tractor on Test : FOTON
 Model : TD904
 Date of Test : January 25, 2013

Table 6. Results of PTO load performance test

SHAFT SPEED (rpm)		PTO		FUEL CONSUMPTION (L/h)	SPECIFIC FUEL CONSUMPTION (g/kW-h)
ENGINE	PTO	TORQUE (kg-m)	POWER (kW)		
2305	617	9.20	5.80	6.18	888.0
2205	601	69.3	42.7	16.2	318.5
2100	580	101.2	60.3	20.9	290.7
2043	561	105.4	60.7	20.1	278.5
1994	541	108.2	60.1	19.7	275.8
1904	521	111.5	59.6	18.8	265.0
1832	499	114.5	58.7	18.9	269.0
1702	462	122.6	58.1	18.3	264.0
1617	432	126.0	55.9	18.4	276.6

REMARKS / OBSERVATIONS: No breakdown and any malfunctioning occurred.

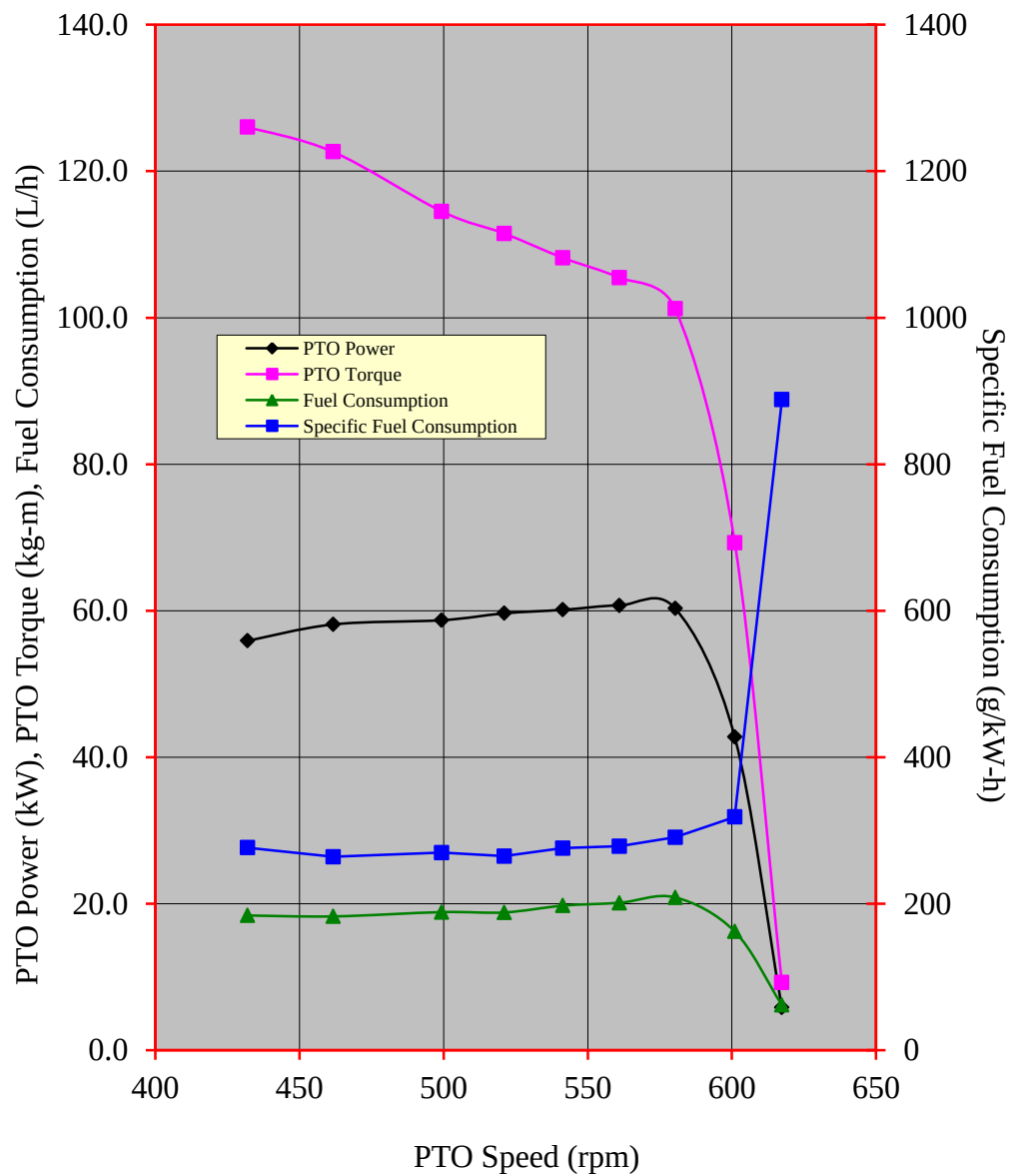


Figure 11. Performance Characteristic Curves of the FOTON TD904 Four-Wheel Tractor

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ROMULO E. EUSEBIO

Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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